

MASTER'S THESIS

Online Imputation of Missing Data in Financial Time Series

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Abstract

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Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. In every one of these settings a practitioner or a researcher faces the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

This question is central for market makers and risk managers. For instance, an emerging markets trader could be asked to make a market for an highly illiquid instrument that has not traded for several hours or days. Depending on the instrument, several approaches are possible: for a bond, she could interpolate the missing yield from the rest of the curve; for a derivative, she could use her own pricing model and hope that the inputs are not too stale.¹ The simplest solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that minimize a loss between the imputed value and the true value, and that preserve as much as possible the statistical properties of the series. In classical machine learning approach, we want to focus on quality of inference rather than interpretability of the model.

Furthermore, and again motivated by practical use cases, we insist on a *causal*

¹Note that here she is faced by the same problem, i.e. factors (as α , β , ρ in SABR) that may have not been observed in a some time.

imputation: the imputed value at time t must be based only on information available up to time t , and not on any future observation. A market maker does not have the luxury of looking into a magical orb to see what the price will be in the next few seconds. We formalise this constraint through the language of stochastic filtrations, survey and taxonomise the algorithms, from classical statistical methods to modern deep generative models, and evaluate them empirically on representative financial datasets.

1.2 The Causal Constraint

Let us quickly formalise basic notation and the notion of causal imputation.

Following Zhengping Che and Liu (2016), we denote a multivariate time series with D variables of length T as $\mathbf{X}_T = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)^\top \in \mathbb{R}^{T \times D}$, where $\mathbf{x}_t \in \mathbb{R}^D$ represents the t -th observations (a.k.a., measurements) of all variables and x_t^d denotes the measurement of d -th variable of $\mathbf{x}_t$. Let $s_t \in \mathbb{R}$ denote the time-stamp when the t -th observation is obtained and we assume that the first observation is made at time $t = 0$ ($s_1 = 0$).

Let $(\Omega, \mathcal{G}, \mathbb{P})$ be a probability space and let $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 1}$ be a filtration representing the information available to a market participant through time. An imputation $\hat{\mathbf{x}}_t^d$ of a missing value $\mathbf{x}_t^d$ is causal if and only if it is $\mathcal{F}_t$ -measurable: it may depend on the history $\{\mathbf{X}_s, \mathbf{M}_s\}_{s \leq t}$ of observed values and missingness indicators up to time t , but on nothing later.

The theoretical benchmark for any causal imputation is the conditional expectation

$$\hat{\mathbf{x}}_t^d = \mathbb{E}[\mathbf{x}_t^d \mid \mathcal{F}_t], \quad (1.1)$$

which is the unique $\mathcal{F}_t$ -measurable estimator minimising mean squared error. In the linear Gaussian state-space model this optimum is attained exactly by the Kalman filter prediction step. For the non-linear, non-Gaussian models that may better describe financial data, it can be approximated by recurrent neural networks, diffusion-based generative models, or transformer architectures.

Note the crucial importance of $\sigma(\mathbf{x}_t)$, the information increment between $\mathcal{F}_{t-1}$ and $\mathcal{F}_t$ generated by the not missing values of $\mathbf{x}_t$ itself.² Without using it, the problem reduces to a purely predictive one, and in my opinion becomes close to impossible: financial markets in a certain sense exist because forecasting the future is impossible or extremely hard, thus the information density of the $\mathcal{F}_{t-1}$ is very low,

² $\sigma(\cdot)$ is a slight abuse of notation since it is not a proper sigma algebra.

and $\hat{\mathbf{x}}_t^d$ will be extremely noisy. On the other hand, the premise of this thesis is that in some datasets $\sigma(\mathbf{x}_t)$ can be informative and useful.

1.3 Use Cases and Datasets

We organise the empirical investigation around six use cases, each representing a distinct type of financial time series and a distinct pattern of missingness.

UC 1 — Tick and high-frequency data (LOBSTER / TAQ). In a limit-order book, quotes are removed when orders are cancelled or modified. The resulting gaps in the bid–ask spread and mid-price are endogenous to price movement: cancellations cluster around volatile episodes, making the mechanism missing not at random (MNAR). The causal constraint is most binding here, since even a sub-second lookahead constitutes inadmissible information for an execution algorithm. We use the LOBSTER dataset, which provides full ten-level limit-order book snapshots for NASDAQ-listed equities under an academic licence, and the Trade and Quote (TAQ) database via WRDS for cross-validation.

UC 2 — Daily equity prices (CRSP / Yahoo Finance). Trading halts triggered by circuit breakers and regulatory suspensions produce gaps of one to five days in the equity price record. Survivorship bias further distorts panels from which delisted stocks have been removed. The missingness is conditionally missing at random (MAR) during ordinary halts but shifts to MNAR when the halt is caused by undisclosed information. We use the Center for Research in Security Prices (CRSP) daily stock file via WRDS as the primary dataset, with Yahoo Finance (yfinance API) as a freely accessible alternative.

UC 3 — Fixed income and yield curves (FRED / ECB SDW). Over-the-counter bond markets produce sparse and irregular quote streams. Off-the-run Treasuries and corporate bonds can go unquoted for days, leaving holes in the yield curve that must be filled causally for real-time risk systems. The missingness pattern ranges from MCAR for the most liquid on-the-run maturities to MNAR for illiquid off-the-run tenors, where the absence of a quote is itself informative about the prevailing liquidity regime. We use the FRED daily yield-curve series (US Treasuries, freely available via the FRED API) and the ECB Statistical Data Warehouse (euro-area sovereign rates) as primary sources.

UC 4 — Macro and mixed-frequency data (FRED-MD). Macro series such as GDP, industrial production, and PMI are published monthly or quarterly, creating structural gaps at the daily frequency required by nowcasting and trading-signal models. Sensor failures and API outages additionally introduce irregular MCAR gaps. The FRED-MD dataset, comprising 128 monthly US macroeconomic series, is widely used as a benchmark in the nowcasting literature, enabling direct comparison with published results.

UC 5 — FX and cryptocurrency (Binance / Dukascopy). Foreign-exchange and cryptocurrency markets operate around the clock, yet liquidity is thin during certain hours or following market stress. The resulting OHLCV gaps are MNAR: illiquidity is endogenous to the bid–ask spread and realised volatility. We use minute-level OHLCV data from the Binance public REST API (cryptocurrency) and tick-level foreign-exchange data from the Dukascopy free historical data service, both of which can be downloaded programmatically without institutional access.

UC 6 — Implied volatility surfaces (CBOE / OptionMetrics). Far out-of-the-money and long-dated options are illiquid; many (strike, maturity) nodes in the implied volatility surface go unquoted on any given day. Imputation here is a two-dimensional problem (strike and maturity) with the additional constraint that any completed surface must respect no-arbitrage conditions: put–call parity and the convexity of option prices as a function of strike. We use the CBOE VIX term-structure series as a freely available proxy and the OptionMetrics IvyDB database via WRDS for the full surface.

The six use cases span all three missingness mechanisms, gap lengths from one tick to several weeks, and data frequencies from millisecond to monthly. For tractability, the main empirical chapters focus on UC 2 (daily equities), UC 3 (yield curves), and UC 5 (high-frequency FX/crypto), which collectively cover the most practically important settings and the widest methodological range. UC 1 and UC 6 are treated as extensions in the final chapter.

1.4 Objectives and Contributions

1.5 Thesis Structure

Chapter 2

Mathematical Foundations

Chapter 3

Literature Review

Chapter 4

Imputation Methods

Chapter 5

Results and Discussion

Chapter 6

Conclusion

6.1 Summary

6.2 Limitations

6.3 Future Work

Bibliography

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